Submodule 33.06: Feature Correlation Matrix & Heatmap Analysis

Topic Overview & Mathematical Formulation

Before performing feature encoding, scaling, and dimensionality reduction, we must analyze the linear interdependencies among our features. The pairwise relationship between two continuous variables X and Y is quantified using the **Pearson Correlation Coefficient (r)**:

$$r_{XY} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

where:

- $r = +1$: Perfect positive linear correlation.
- $r = 0$: No linear relationship (orthogonal features).
- $r = -1$: Perfect negative linear correlation.

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1. Python Implementation

Reference: smartcart.ipynb (Cells 24 - 28).

```
import matplotlib.pyplot as plt
import seaborn as sns

# Compute Pearson correlation matrix on numeric features
corr = df_cleaned.corr(numeric_only=True)

# Generate heatmap visualization
plt.figure(figsize=(8, 6))
sns.heatmap(
    corr,
    annot=True,
    annot_kws={"size": 6},
    cmap="coolwarm"
)
plt.title("Feature Correlation Heatmap")
plt.show()
```

2. In-Depth Correlation Analysis & Business Findings

Feature Pair	Correlation (r)	Direction	Business Interpretation
`Income` vs `Total_Spending`	**+0.79**	Strong Positive	Strongest economic driver. High household earnings directly translate to high consumer expenditure.
`Total_Spending` vs `NumCatalogPurchases`	**+0.78**	Strong Positive	High spenders heavily utilize premium curated product catalogs.
`Income` vs `NumCatalogPurchases`	**+0.69**	Moderate Positive	Affluent customers strongly favor mail/digital catalog channels.
`Total_Spending` vs `NumStorePurchases`	**+0.67**	Moderate Positive	High spenders frequently shop in physical retail stores.
`Income` vs `NumStorePurchases`	**+0.63**	Moderate Positive	Higher income correlates with higher in-person store visits.
`Income` vs `NumWebVisitsMonth`	**-0.65**	Strong Negative	High earners visit the e-commerce website less frequently; low-income customers browse frequently.
`NumStorePurchases` vs `NumWebVisitsMonth`	**-0.43**	Moderate Negative	Offline brick-and-mortar shoppers do not spend time browsing online web pages.
`Customer_Tenure_Days`, `Complain`, `Response`	≈ 0.0	Neutral / Orthogonal	These attributes capture independent dimensions of customer behavior (loyalty duration and friction).

3. Methodological Takeaways

1. High multicollinearity between `Income`, `Total_Spending`, and `NumCatalogPurchases` confirms that financial capacity is the primary variance axis in this dataset.
2. The strong negative relationship between income and website visits indicates clear behavioral dichotomy: **Affluent Spenders (low browsing, direct high-value purchases)** vs **Budget Browsers (high browsing, low purchase conversion)**.
3. This strong collinearity provides ideal justification for **Principal Component Analysis (PCA)** to condense these correlated axes into orthogonal principal components.